

Experiment No.2

Title: Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

Software/Tools Required:

- Visual Studio Code (or any text editor)
- Web browser (Chrome, Edge, Firefox, etc.) to render the HTML page
- Basic knowledge of HTML5 and modern JavaScript (ES6+) and CSS.

Theory:

- JavaScript provides different ways to declare variables, each with its own scope and purpose.
 1. var: Function-scoped and can be re-declared. Its use is mostly avoided in modern JavaScript due to unpredictable scoping behavior.
- let: Block-scoped and allows reassignment. It is best used for variables whose values will change over time (like loop counters or running totals).
- const: Block-scoped and cannot be reassigned. It is used for values that must remain constant throughout the script, ensuring data integrity.

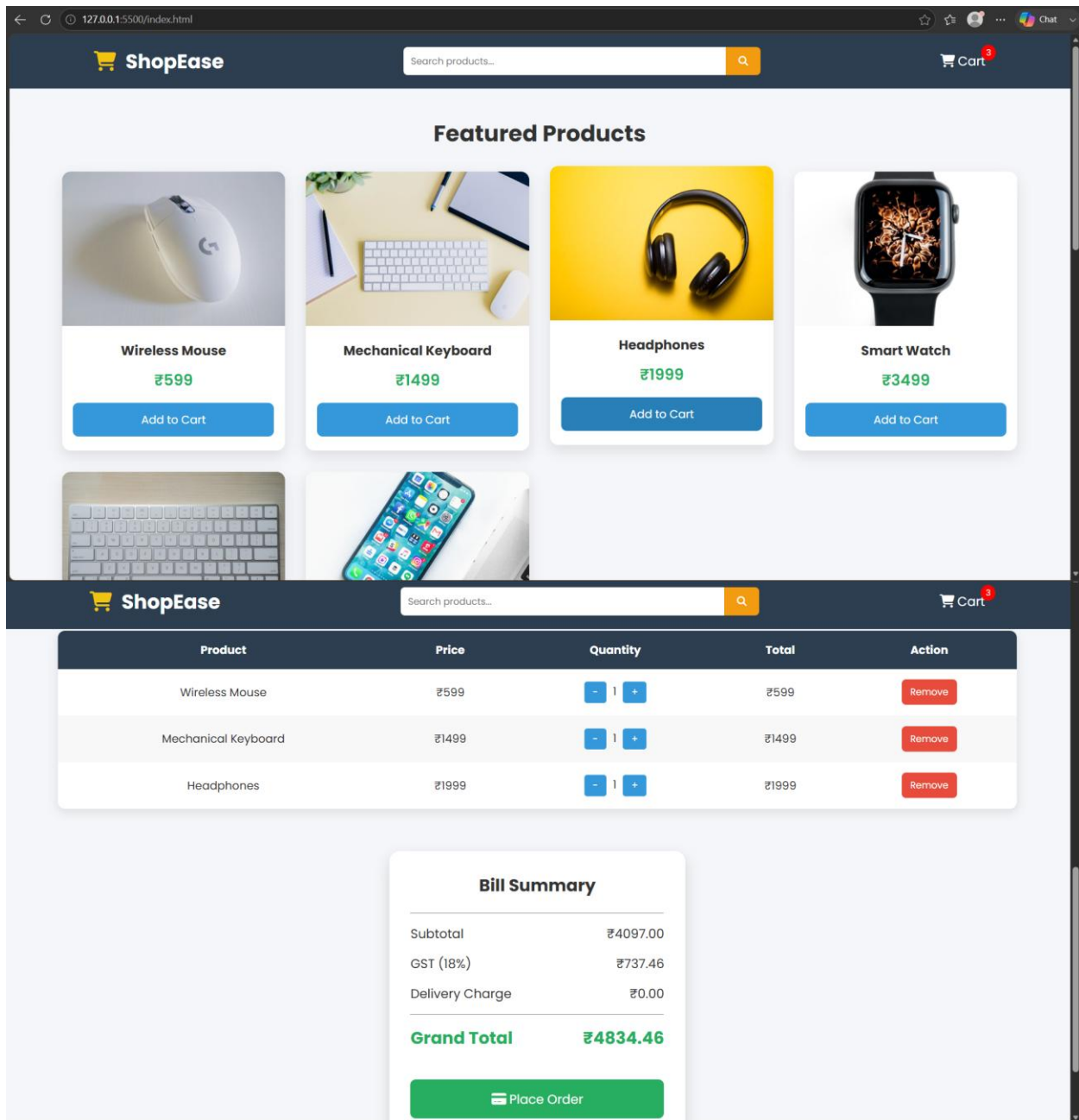
Modern ES6 Features:

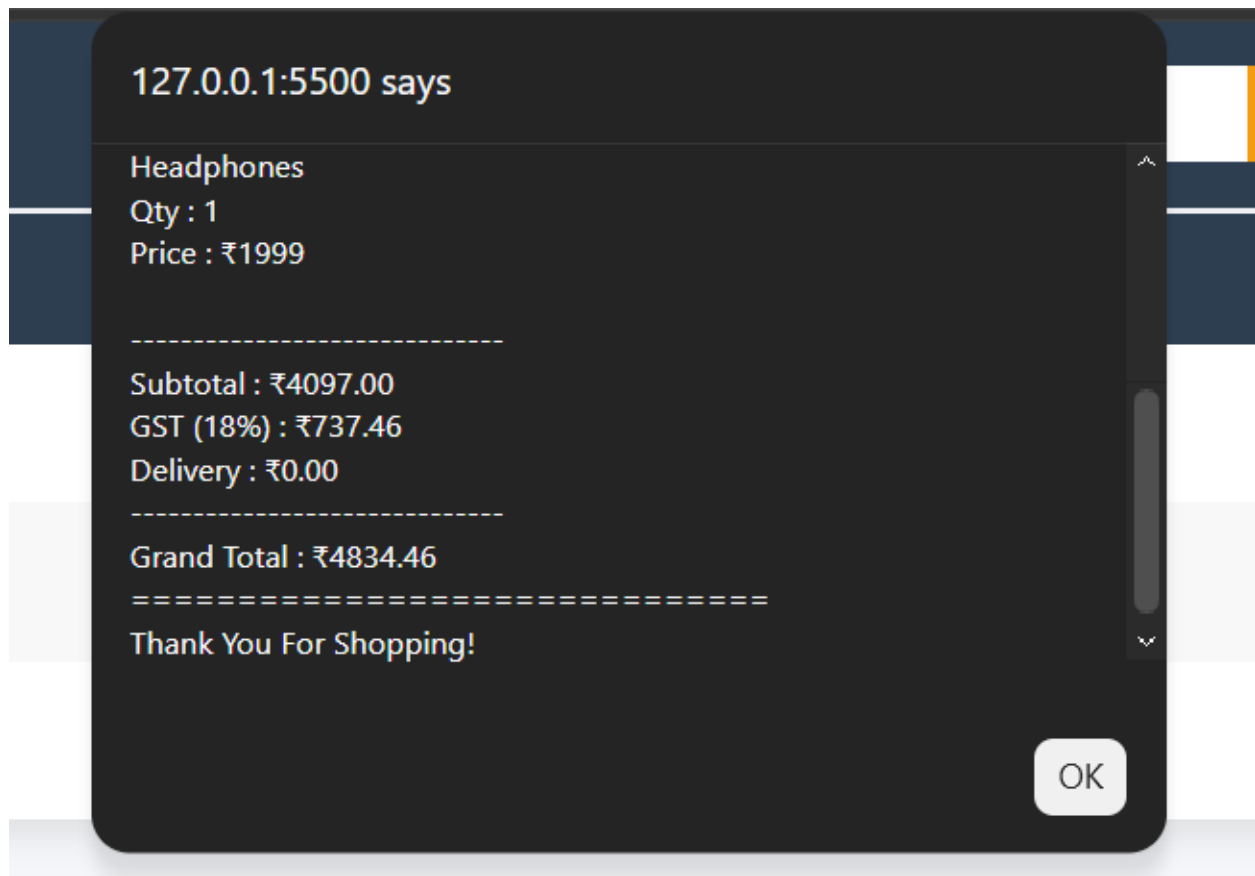
- Template Literals: Created using backticks (` `), they allow for multi-line strings and seamless embedding of variables and expressions directly into text using the `${expression}` syntax.
- Destructuring: A convenient syntax that extracts values from arrays or properties from objects into distinct, individual variables, leading to cleaner and more readable code.

Aim of the CASE STUDY: This experiment demonstrates the use of var, let, const, template literals and object destructuring by developing a case study of simple Online Shopping Website. The webpage accepts customer details and product information, processes the purchase details, calculates the total amount and GST, and dynamically displays an order summary. Template literals are used to generate the order details, while destructuring is used to extract product information efficiently.

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File Path: "C:\Users\ASUS\Desktop\JavaScript Lab"





Output:

On opening the HTML file in a browser, a "Billing Calculator" page is displayed. Upon entering the required data, a dynamically generated card titled "Your Bill" appears on the screen. As captured in the screenshot, the receipt successfully extracts and displays the Customer Name (Rugved Madankar) and Product (Keyboard, Mouse, Headphones). It accurately processes the large numerical inputs (Quantity, Price per Unit) to calculate the Total Amount, computes the 18% GST, and outputs the Final Amount.

Result / Conclusion:

The experiment was successfully completed. The concepts of var, let, and const were implemented to understand variable declaration and scope. Modern JavaScript features such as template literals and destructuring were successfully used while developing an Online Shopping Website. The application accepted user input, processed product and customer details, calculated the total amount and GST, and displayed a dynamically generated order summary, demonstrating the practical implementation of ES6 features in web development and prints a bill at last.